Collage reconstruction: a tightened proof

Bisets over endomorphism monoids, the tensor coequalizer, and the reconstruction of a small category

MacBeth
Kodamai

10 June 2026

Abstract

This note replaces the hand-waved proof of the "collage reconstruction" theorem (part (1) of the Corrected Decomposition theorem in the two-atoms / Zappa–Szép note) by a fully element-wise argument. We define the tensor product of bisets over a monoid as an explicit coequalizer, prove that composition in a small category is balanced and therefore descends to the tensor, verify the category axioms on representatives, and prove that the collage construction and its converse are mutually inverse. One genuine subtlety in the converse — the need for the regular-biset embedding of each vertex monoid to be *strictly* compatible with composition — is isolated and discharged explicitly rather than glossed.

1 The tensor product of bisets over a monoid

Throughout, M, N, P are monoids (written multiplicatively, units 1). A *right M-set* is a set X with a map $X \times M \to X$, $(x, m) \mapsto x \cdot m$, satisfying $x \cdot 1 = x$ and $(x \cdot m) \cdot m' = x \cdot (mm')$. Left M-sets are dual. An (N, M)-*biset* is a set carrying a left N-action and a right M-action that commute: $(n \cdot x) \cdot m = n \cdot (x \cdot m)$.

Definition 1 (Tensor over a monoid as a coequalizer)**.** Let X be a right M-set and Y a left M-set. Consider the two maps

$$\lambda, \rho : X \times M \times Y \ \rightrightarrows\ X \times Y, \qquad \lambda(x, m, y) = (x \cdot m,\ y), \quad \rho(x, m, y) = (x,\ m \cdot y).$$

The *tensor product* $X \otimes_M Y$ is the coequalizer of λ, ρ in **Set**: the quotient $(X \times Y)/{\sim}$, where $\sim$ is the smallest equivalence relation with

$$(x \cdot m,\ y)\ \sim\ (x,\ m \cdot y) \qquad \text{for all } x \in X,\ m \in M,\ y \in Y.$$

We write $x \otimes y$ for the class of (x, y) and $q : X \times Y \to X \otimes_M Y$ for the quotient map. By construction $x \cdot m \otimes y = x \otimes m \cdot y$.

Remark 2. The generating relation is already reflexive ($m = 1$), but it is neither symmetric nor transitive in general, so $\sim$ is its reflexive–symmetric–transitive closure: $(x, y) \sim (x', y')$ iff there is a finite zig-zag

$$(x, y) = (x_0, y_0)\ \leftrightarrow\ (x_1, y_1)\ \leftrightarrow\ \cdots\ \leftrightarrow\ (x_k, y_k) = (x', y'),$$

each step $\leftrightarrow$ being a generator applied left-to-right or right-to-left. We will need this explicit zig-zag description when checking that maps out of the tensor are well defined.

The universal property is the defining one for a coequalizer; we record the form we use.

Lemma 3 (Universal property / balance). *Let X be a right M-set, Y a left M-set, Z any set. A map $\beta : X \times Y \to Z$* descends *to a (necessarily unique) map $\bar{\beta} : X \otimes_M Y \to Z$ with $\bar{\beta}(x \otimes y) = \beta(x, y)$ if and only if β is* balanced*:*

$$\beta(x \cdot m,\, y) = \beta(x,\, m \cdot y) \qquad \textit{for all } x \in X,\ m \in M,\ y \in Y. \tag{B}$$

Proof. ($\Rightarrow$) If $\bar{\beta}$ exists then $\beta(x \cdot m, y) = \bar{\beta}(x \cdot m \otimes y) = \bar{\beta}(x \otimes m \cdot y) = \beta(x, m \cdot y)$.

($\Leftarrow$) Suppose (B). Then β is constant on each generator pair, hence (being a function landing in a set, so respecting symmetry and composing along the zig-zag of Remark 2) constant on each $\sim$-class: if $(x, y) \leftrightarrow (x', y')$ is a single generator step in either direction, (B) gives $\beta(x, y) = \beta(x', y')$, and chaining over the finite zig-zag gives equality on the whole class. Therefore $\bar{\beta}([x, y]) := \beta(x, y)$ is well defined; it satisfies $\bar{\beta} \circ q = \beta$, and uniqueness is forced because q is surjective. $\square$

2 The collage of a small category

Fix a small category $\mathcal{C}$. For objects a, b write $\mathrm{Hom}(a, b)$ for the hom-set and $\mathrm{End}(a) := \mathrm{Hom}(a, a)$ for the endomorphism monoid (composition, unit id_a). As in the source note, $\mathrm{Hom}(a, b)$ is canonically an $(\mathrm{End}(b), \mathrm{End}(a))$-biset: the right $\mathrm{End}(a)$-action is precomposition $f \cdot e := f \circ e$ and the left $\mathrm{End}(b)$-action is postcomposition $e' \cdot f := e' \circ f$; the two commute by associativity, so the biset axiom holds.

(i) Balance

Proposition 4 (Composition is balanced). *Fix objects a, b, c. The composition map*

$$\gamma : \mathrm{Hom}(b, c) \times \mathrm{Hom}(a, b) \to \mathrm{Hom}(a, c), \qquad \gamma(g, f) = g \circ f,$$

is $\mathrm{End}(b)$*-balanced: for all $g \in \mathrm{Hom}(b, c)$, $e \in \mathrm{End}(b)$, $f \in \mathrm{Hom}(a, b)$,*

$$\gamma(g \cdot e,\, f) = \gamma(g,\, e \cdot f), \qquad \textit{i.e.} \qquad (g \cdot e) \circ f = g \circ (e \cdot f).$$

Hence by Lemma 3 it descends to a unique map

$$\mu_{a,b,c} : \mathrm{Hom}(b, c) \otimes_{\mathrm{End}(b)} \mathrm{Hom}(a, b) \longrightarrow \mathrm{Hom}(a, c), \qquad \mu_{a,b,c}(g \otimes f) = g \circ f.$$

Proof. Here $g \cdot e = g \circ e$ (right $\mathrm{End}(b)$-action on $\mathrm{Hom}(b, c)$) and $e \cdot f = e \circ f$ (left $\mathrm{End}(b)$-action on $\mathrm{Hom}(a, b)$). Associativity of composition in $\mathcal{C}$ gives

$$(g \cdot e) \circ f = (g \circ e) \circ f = g \circ (e \circ f) = g \circ (e \cdot f),$$

which is exactly condition (B) of Lemma 3 for $\beta = \gamma$ and $M = \mathrm{End}(b)$. The descended map $\mu_{a,b,c}$ and its formula are then immediate from the lemma. $\square$

(ii) Reconstruction

Definition 5 (The collage category $\mathcal{C}^\flat$). Define a structure $\mathcal{C}^\flat$ with

- objects: $\mathrm{ob}\, \mathcal{C}$;
- hom-sets: $\mathcal{C}^\flat(a, b) := \mathrm{Hom}(a, b)$ (the underlying set of the biset);

- composition: $\mu_{a,b,c}$ of Proposition 4, precomposed with the quotient q, i.e. $g \circ^\flat f := \mu_{a,b,c}(g \otimes f)$;
- identities: $\mathrm{id}_a^\flat := 1_{\mathrm{End}(a)} = \mathrm{id}_a$, the monoid unit.

Theorem 6 (Reconstruction)**.** *$\mathcal{C}^\flat$ is a category, and the object-identity assignment $F : \mathcal{C} \to \mathcal{C}^\flat$, $F(a) = a$ on objects and $F(f) = f$ on morphisms, is an isomorphism of categories with inverse the identity-on-everything map.*

Proof. We verify the axioms element-wise; since $\mathcal{C}^\flat(a,b) = \mathrm{Hom}(a,b)$ as sets and, by the formula in Proposition 4,

$$g \circ^\flat f \;=\; \mu_{a,b,c}(g \otimes f) \;=\; g \circ f,$$

composition in $\mathcal{C}^\flat$ is literally composition in $\mathcal{C}$, and $\mathrm{id}_a^\flat = \mathrm{id}_a$. Thus every axiom of $\mathcal{C}^\flat$ is the corresponding axiom of $\mathcal{C}$:

Well-definedness. $\mu_{a,b,c}$ is a genuine function on the tensor by Proposition 4, and q is a function, so $\circ^\flat$ is a well-defined map $\mathrm{Hom}(b,c) \times \mathrm{Hom}(a,b) \to \mathrm{Hom}(a,c)$.

Left unit. $\mathrm{id}_b^\flat \circ^\flat f = \mathrm{id}_b \circ f = f$ for $f \in \mathrm{Hom}(a,b)$.

Right unit. $f \circ^\flat \mathrm{id}_a^\flat = f \circ \mathrm{id}_a = f$.

Associativity. For $f \in \mathrm{Hom}(a,b)$, $g \in \mathrm{Hom}(b,c)$, $h \in \mathrm{Hom}(c,d)$,

$$(h \circ^\flat g) \circ^\flat f = (h \circ g) \circ f = h \circ (g \circ f) = h \circ^\flat (g \circ^\flat f),$$

by associativity in $\mathcal{C}$.

Hence $\mathcal{C}^\flat$ is a category. The map F is the identity on objects and on each hom-set, and it preserves composition and identities because $g \circ^\flat f = g \circ f$ and $\mathrm{id}_a^\flat = \mathrm{id}_a$. A bijective-on-objects, bijective-on-homs functor is an isomorphism of categories, with the set-theoretic identity as inverse functor. ☐

Remark 7 (Where the tensor actually does work)**.** The tensor $\otimes_{\mathrm{End}(b)}$ is not cosmetic: it is exactly the data needed to phrase composition without reference to the ambient category, by quotienting the formal pair-set $\mathrm{Hom}(b,c) \times \mathrm{Hom}(a,b)$ by the $\mathrm{End}(b)$-loop relation. The non-trivial content of Proposition 4 is that the relation imposed by the coequalizer is *respected* by composition; once that holds, μ exists and the category axioms transfer verbatim. In particular the often mysterious "associativity is compatibility of the tensor glueing" reduces, on representatives, to plain associativity in $\mathcal{C}$ — because $\mu(g \otimes f)$ is computed by a chosen representative and the value $g \circ f$ is representative-independent by balance.

(iii) Converse: from a collage datum to a category

We now axiomatize the data abstractly and show the construction inverts.

Definition 8 (Biset-collage datum)**.** A *biset-collage datum* $\mathcal{D}$ consists of:

1. a set O of objects;
2. for each $x \in O$ a monoid M_x (the vertex monoid), with unit 1_x;

3. for each ordered pair (a,b) an (M_b, M_a)-biset $H(a,b)$, with a *distinguished* biset isomorphism $\iota_x : M_x \xrightarrow{\sim} H(x,x)$ for $a = b = x$, where M_x is viewed as the regular (M_x, M_x)-biset (left and right action = multiplication); we write $\underline{e} := \iota_x(e) \in H(x,x)$ and identify M_x with its image, so that $\underline{1_x} \in H(x,x)$ is a distinguished element and the biset actions of M_x on every $H(a,b)$ agree with multiplication on $H(x,x)$ in the sense of (U) below;

4. for each triple (a,b,c) a *balanced* composition

$$c_{a,b,c} : H(b,c) \times H(a,b) \to H(a,c), \qquad c(g, e \cdot f) = c(g \cdot e, f) \quad (e \in M_b),$$

inducing $m_{a,b,c} : H(b,c) \otimes_{M_b} H(a,b) \to H(a,c)$ by Lemma 3;

subject to the axioms (writing $g \bullet f := c(g,f)$):

(U) Unit. For $f \in H(a,b)$, $\underline{1_b} \bullet f = f = f \bullet \underline{1_a}$, and moreover the distinguished left/right actions are recovered by composition with elements of the vertex monoids: for $e \in M_b$, $\underline{e} \bullet f = e \cdot f$, and for $e \in M_a$, $f \bullet \underline{e} = f \cdot e$. (Equivalently: on $H(b,b) \times H(a,b)$, $\bullet$ restricts to the left M_b-action under ι_b, and dually on the right.)

(A) Associativity. For $f \in H(a,b)$, $g \in H(b,c)$, $h \in H(c,d)$, $(h \bullet g) \bullet f = h \bullet (g \bullet f)$.

Theorem 9 (Converse and mutual inverse)**.**

1. (Reconstruction.) *Every biset-collage datum $\mathcal{D}$ yields a small category $\mathcal{C}_\mathcal{D}$ with* $\mathrm{ob}\,\mathcal{C}_\mathcal{D} = O$*,* $\mathcal{C}_\mathcal{D}(a,b) = H(a,b)$*, composition $g \circ f := g \bullet f$, and identities* $\mathrm{id}_a := \underline{1_a}$*.*

2. (Mutual inverse.) *The two constructions are inverse up to isomorphism: $(\mathcal{C}^\flat)$-of-$\mathcal{D}$ on the collage datum extracted from a category returns that category (Theorem 6), and conversely $\mathcal{C}_{\mathcal{D}(\mathcal{C}_\mathcal{D})} = \mathcal{C}_\mathcal{D}$ on the nose, where $\mathcal{D}(\mathcal{C}_\mathcal{D})$ is the collage datum extracted from $\mathcal{C}_\mathcal{D}$ by the recipe of Section 2 ($M_x = \mathrm{End}_{\mathcal{C}_\mathcal{D}}(x)$, $H = \mathrm{Hom}_{\mathcal{C}_\mathcal{D}}$, $\iota_x = \mathrm{id}$).*

Proof. (1) Axioms (U) and (A) are precisely the unit and associativity laws of a category, stated for $\bullet$, with $\underline{1_a}$ as identity at a. Well-definedness of composition as a map $H(b,c) \times H(a,b) \to H(a,c)$ is part of the datum (item 4); balance is not even needed for $\mathcal{C}_\mathcal{D}$ to be a category, but it guarantees that $\bullet$ factors through $H(b,c) \otimes_{M_b} H(a,b)$, which is what makes $\mathcal{C}_\mathcal{D}$ a genuine collage rather than an arbitrary composition law. Smallness holds because O and all $H(a,b)$ are sets.

(2) *Category $\to$ datum $\to$ category.* Starting from a small category $\mathcal{C}$, the extracted datum has $O = \mathrm{ob}\,\mathcal{C}$, $M_x = \mathrm{End}(x)$, $H(a,b) = \mathrm{Hom}(a,b)$ with the canonical biset structure, $\iota_x = \mathrm{id}$ (the regular biset of $\mathrm{End}(x)$ *is* $\mathrm{Hom}(x,x)$ with its pre/post-composition actions), and $c_{a,b,c} = \gamma$ (composition), which is balanced by Proposition 4. Axiom (U) for this datum says $\mathrm{id}_b \circ f = f = f \circ \mathrm{id}_a$ and $\underline{e} \bullet f = e \circ f = e \cdot f$: the first is the unit law of $\mathcal{C}$, the second is the definition of the left action; both hold. Axiom (A) is associativity in $\mathcal{C}$. Thus the extracted datum is a legitimate biset-collage datum, and $\mathcal{C}_{\mathcal{D}(\mathcal{C})}$ has the same objects, the same hom-sets, composition $g \bullet f = g \circ f$ and identities $\underline{1_a} = \mathrm{id}_a$ — i.e. $\mathcal{C}_{\mathcal{D}(\mathcal{C})} = \mathcal{C}$ on the nose. (This is the content of Theorem 6 read backwards.)

Datum $\to$ category $\to$ datum. Starting from a datum $\mathcal{D}$, form $\mathcal{C}_\mathcal{D}$ as in (1); then extract a datum $\mathcal{D}' := \mathcal{D}(\mathcal{C}_\mathcal{D})$. We check $\mathcal{D}' = \mathcal{D}$. Objects: $\mathrm{ob}\,\mathcal{C}_\mathcal{D} = O$. Vertex monoids: $\mathrm{End}_{\mathcal{C}_\mathcal{D}}(x) = H(x,x)$ as a *set*, with multiplication $g \bullet f$; by axiom (U) restricted to $x = a = b = c$, for $\underline{e}, \underline{e'} \in \iota_x(M_x)$ we have $\underline{e'} \bullet \underline{e} = e' \cdot \underline{e} = \underline{e'e}$ (left action = multiplication via ι_x, since ι_x is a biset iso onto the regular biset). Because $\iota_x : M_x \to H(x,x)$ is a bijection, every element of $H(x,x)$ is some $\underline{e}$, so multiplication in $\mathrm{End}_{\mathcal{C}_\mathcal{D}}(x)$ is $\underline{e'} \bullet \underline{e} = \underline{e'e}$, i.e. ι_x is a monoid isomorphism $M_x \cong \mathrm{End}_{\mathcal{C}_\mathcal{D}}(x)$; identifying along ι_x (as the datum already does), $M_x = \mathrm{End}_{\mathcal{C}_\mathcal{D}}(x)$ with equal units $\underline{1_x} = \mathrm{id}_x$. Hom-sets: $H(a,b) = \mathrm{Hom}_{\mathcal{C}_\mathcal{D}}(a,b)$ by definition. Biset actions: the extracted left $\mathrm{End}_{\mathcal{C}_\mathcal{D}}(b)$-action on $\mathrm{Hom}_{\mathcal{C}_\mathcal{D}}(a,b)$ is postcomposition $\underline{e} \bullet f$, which by axiom (U) equals the original left action $e \cdot f$; dually on the right. Hence the extracted biset structure equals the original. Composition: $\bullet$ is unchanged. Therefore $\mathcal{D}' = \mathcal{D}$, as required. □

Remark 10 (The one genuine subtlety, isolated)**.** The single place where strictness could fail in the converse is the clause "ι_x is a biset isomorphism onto the *regular* biset" together with axiom (U)'s identity $\underline{e} \bullet f = e \cdot f$. Without it, the distinguished elements $\underline{e} \in H(x,x)$ would compose by some abstract $\bullet$ that need not coincide with the monoid multiplication of M_x, and then "$\mathrm{End}_{\mathcal{C}_\mathcal{D}}(x) = M_x$" would hold only up to a chosen isomorphism, not on the nose. We have made this compatibility *part of the datum* (items 3–4 and axiom (U)); under it the round trip is an equality. If one weakens the datum to allow ι_x to be merely a monoid *homomorphism* that is bijective on the underlying set but *not* required to intertwine $\bullet$ with M_x-multiplication, the two constructions are inverse only up to canonical isomorphism of categories, not equal. We flag this explicitly: the "mutually inverse on the nose" statement is true exactly for data satisfying the regular-embedding axiom (U), and is the natural normal form; for general data one gets an equivalence (indeed isomorphism) of categories rather than an identity. No gap remains in the stated (strict) version.

3 Summary of the logical skeleton

1. **Tensor** = coequalizer of the two action maps (Definition 1); maps out of it = balanced maps (Lemma 3).
2. **Balance** of composition = associativity in $\mathcal{C}$ (Proposition 4); hence $\mu(g \otimes f) = g \circ f$ is well defined.
3. **Reconstruction:** $\mathcal{C}^\flat$ has composition literally equal to that of $\mathcal{C}$, so the category axioms transfer verbatim and $\mathcal{C} \cong \mathcal{C}^\flat$ (Theorem 6).
4. **Converse:** a biset-collage datum with the regular-embedding axiom (U) and associativity (A) is exactly a small category, and the two constructions are mutually inverse on the nose (Theorem 9), the only subtlety being the strict compatibility of the vertex-monoid embedding, which is isolated in Remark 10.